



Karina A. Aquino

• doctoral candidate • geologist • researcher •
born 24 September 1992
Filipino

CONTACT



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scholar.google.com/citations?user=FSjyZRAAAAAJ

LANGUAGES

Tagalog	<i>Mother tongue</i>
English	<i>Proficient</i>
German	<i>Conversational</i>
Hungarian	<i>Basic</i>

ABOUT ME

My main motivation for doing the PhD is to be able to return to the Philippines and teach and do some research.

I am currently using mineralogy and geochemistry (e.g., stable isotope geochemistry) in understanding fluid-rock-microbe interactions in low temperature, serpentinization-influenced hydrothermal settings.

EDUCATION

- **Doctoral candidate in Geology; 2018-present**
expected graduation: December 2022
Department of Earth Sciences, ETH Zürich (Swiss Federal Institute of Technology), Switzerland
Dissertation: *Mineral-microbe-fluid interactions at the Lost City hydrothermal field*
- **Master of Science in Geology; 2013-2018**
National Institute of Geological Sciences, University of the Philippines (UP NIGS), Diliman, Quezon City
Dissertation: *Spatio-temporal evolution of laterization: Insights from detailed mineralogical characterization and reactive transport modelling of Sta. Cruz nickel laterite deposit, Zambales, Philippines*
- **Bachelor of Science in Geology; 2009-2013**
National Institute of Geological Sciences, University of the Philippines (UP NIGS), Diliman, Quezon City
Research: *Rediscovering Antipolo Basalt: Characterization and Correlation using Petrography and Geochemistry.*
graduated *magna cum laude*
- **High school; 2005-2009**
Philippine Science High School, Main Campus, Diliman, Quezon City
graduated *with high honors*

AWARDS & GRANTS

2018-2022	Department of Science and Technology – Science Education Institute (DOST-SEI) Foreign graduate fellowship
2016	Best student poster, <i>Paladutaan</i> : UP NIGS Research Conference
2016	University of the Philippines Office of the Vice President for Academic Affairs short-term training grant for faculty
2016	Dr. Deane K. Smith Tuition Waiver Grant, International Center for Diffraction Data (ICDD)
2015	Justice Feliciano Graduate fellowship
2014	Philippine Geology Licensure examination, Rank 6
2014	Joker Arroyo award
2013	University of the Philippines Geology Merit award
2013	Philippine National Oil Company -Exploration Corporation (PNOC-EC) undergraduate scholarship
2010-2013	DOST-SEI undergraduate scholarship
2010-2013	University of the Philippines Presidential scholarship

PROFESSIONAL EXPERIENCE

- **Mentor, GradMap Philippines; 2020-2021(2 cycles)**
Involved in mentoring aspiring Filipino STEM students through their graduate school application process; two successful mentees accepted at ETH Zürich
- **Participant, International Ocean/Continental Drilling Program (IODP/ICDP) New Caledonian Peridotite Amphibious Drilling Workshop, Montpellier, France; 2019**
Workshop that aims at developing an amphibious IODP/ICDP proposal for drilling the onshore peridotites of New Caledonia and their offshore extension
- **Instructor 4, UP NIGS; 2013-2018**
Classes taught: General Geology (lecture); Elementary Petrology (laboratory); Elementary Crystallography and Mineralogy (lecture & laboratory); Field techniques in Geology; Metalliferous ore deposits (laboratory).
Main contributor to the new Geology 41 (core course) and Geology 141 (elective) course outlines for the new curriculum in response to the Kto12 education program.
Significant contributions to the Geology 40, 50, and 194 laboratory manuals.
- **Guest student, Woods Hole Oceanographic Institution, Massachusetts, USA; 2016**
Invited by Dr. Frieder Klein, Marine Chemistry and Geochemistry Department; Analysis of olivine hosted fluid inclusions via confocal Raman spectroscopy

- Participant, XRD Clinics, International Center for Diffraction Data, Newtown Square, PA; 2016
Fundamentals of X-ray Powder Diffraction, May 16 to 20, 2016
Advanced Methods in X-ray Powder Diffraction, May 23 to 27, 2016
- Intern, Hokkaido University, Japan; 2014
Used ion chromatograph and inductively coupled plasma atomic emission spectrometer for fluoride adsorption study; Research project: *Immobilization of fluoride leached from tunnel excavated rocks using natural adsorbents*
- Student assistant, Earth Materials Sciences Laboratory, UP NIGS; 2011 - 2013
Performed characterization of materials using X-ray Diffraction (XRD), X-ray Fluorescence Spectrometry (XRF) and Fourier Transform Infrared Spectroscopy (FTIR); Assisted in a geomicrobiological fieldwork in various hyperalkaline environments in the Philippines

PUBLICATIONS

- [1] Brazelton, W.J., McGonigle, J.M., Motamedi, S., Pendleton, H.L., Twing, K.I., Miller, B.C., Lowe, W.J., Hoffman, A.M., Prator, C.A., Chadwick, G.L., Anderson, R.E., Thomas, E., Butterfield, D.A., **Aquino, K.A.**, Früh-green, G.L., Schrenk, M.O., Lang, S.Q. (2022). Metabolic strategies shared by basement residents of the Lost City hydrothermal field. *Applied and Environmental Microbiology*.
- [2] **Aquino, K.A.**, Früh-Green, G.L., Rickli, J., Bernasconi, S.M., Lang, S.Q., Lilley, M.D., & Butterfield, D.A. (2022). Multi-stage evolution of the Lost City hydrothermal vent fluids. *Geochimica et Cosmochimica Acta*.
<https://doi.org/10.1016/j.gca.2022.06.027>
- [3] **Aquino, K.A.**; Arcilla, C.A.; Schardt, C.; Tupaz, C.A.J. (2022). Mineralogical and Geochemical Characterization of the Sta. Cruz Nickel Laterite Deposit, Zambales, Philippines. *Minerals*, 12, 305.
<https://doi.org/10.3390/min12030305>
- [4] **Aquino, K.A.**; Arcilla, C.A.; Schardt, C.; Tupaz, C.A.J. (2022). Serpentinization and Weathering of Peridotite: A Study on the Mineralogical and Geochemical Evolution of the Sta. Cruz Nickel Laterite Deposit, Zambales, Philippines. <https://www.preprints.org/manuscript/202112.0155/v2>
- [5] **Aquino, K.A.**; Arcilla, C.A.; Schardt, C.S. (2019). Mineralogical Zonation of the Sta. Cruz Nickel Laterite Deposit, Zambales, Philippines Obtained from Detailed X-ray Diffraction Coupled with Rietveld Refinement. *Journal of the Geological Society of the Philippines*, 73, 1–14

CONFERENCE ABSTRACTS

- [1] **Aquino, K.A.**, Früh-Green, G.L., Bernasconi, S.M., Foubert, A., Bontognali, T., Lang, S.Q. (2022) Brucite biomineralization at the Lost City hydrothermal field, 30°N MAR. 12th International Symposium Geochemistry of the Earth's Surface (GES12). Zürich, Switzerland. [poster]
- [2] **Aquino, K.A.**, Früh-Green, G.L., Rickli, J., Foubert, A., Bernasconi, S.M., Lang, S.Q. (2022) Fluid mixing and spatial mineralogical and geochemical variability in the Lost City hydrothermal field chimneys. 32nd Goldschmidt Conference (Goldschmidt2022). Honolulu, Hawai'i and online. [oral presentation]
- [3] **Aquino, K.A.**, Früh-Green, G.L., Bernasconi, S.M., Lang, S.Q. (2020) Multiple processes controlling vent fluid compositions in the Lost City hydrothermal field. International Conference on Ophiolites and the Oceanic Lithosphere: Results of the Oman Drilling Project and Related Research, Sultan Qaboos University, Muscat, Sultanate of Oman. [poster]
- [4] **Aquino, K.A.**, Früh-Green, G.L., Bernasconi, S.M., Lang, S.Q. (2019) Stable isotope geochemistry of carbonate-brucite chimneys and vent fluids from Lost City hydrothermal field, 30°N MAR. 29th Goldschmidt Conference (Goldschmidt2019). Barcelona, Spain. [poster]
- [5] **Aquino, K.A.**, Arcilla, C.A., Schardt, C. (2019). Serpentinization and weathering of peridotite from the Zambales Ophiolite, Philippines. New Caledonia Peridotite Amphibious Drilling Workshop, Montpellier, France. [poster]
- [6] **Aquino, K.A.**, Arcilla, C.A., Schardt, C. (2018). Serpentinization and weathering of peridotite from the Zambales Ophiolite, Philippines. The Royal Society Serpentinite in the Earth System Discussion Meeting. London, United Kingdom. [poster]
- [7] **Aquino, K.A.**, Arcilla, C.A., Schardt, C. (2018). Mineralogical and geochemical evolution of laterization: a study of the weathering mantle of the Zambales Ophiolite Complex, Philippines. 15th Annual Meeting of Asia Oceania Geosciences Society (AOGS2018). Honolulu, Hawaii. [oral presentation]
- [8] Cosalan, P.A., **Aquino, K.A.**, Arcilla, C.A. Geochemical Analysis of the Zambales Nickel Laterite Deposit, Philippines. 15th Annual Meeting of Asia Oceania Geosciences Society (AOGS2018). Honolulu, Hawaii. [poster]
- [9] **Aquino, K.A.**, Arcilla, C.A., Schardt, C. (2017). Spatio-temporal evolution of laterization: Insights from mineralogical characterization and reactive transport modelling of Zambales nickel laterite deposit, Philippines. 27th Goldschmidt Conference (Goldschmidt2017). Paris, France. [poster]
- [10] **Aquino, K.A.**, Arcilla, C.A. (2016). Geochemical Modelling of a Nickel Laterite Deposit: Effect of Degree of Serpentinization on Ultramafic Weathering. 26th Goldschmidt Conference (Goldschmidt2016). Yokohama, Japan. [poster]

- [11] **Aquino, K.A.**, Codillo, E.A., Arcilla, C.A. (2015). Ultramafic Weathering and Laterization: Reactive Transport Geochemical Modelling of a Philippine Iron-Nickel Laterite. 25th Goldschmidt Conference (Goldschmidt2015). Prague, CZ. [poster]
- [12] **Aquino, K.A.**, Tarriela, M. B., Lagata, C.L., (2015). Mineral Equilibrium in the Hydrothermal Waters of the Alto Peak Geothermal System. 12th Annual Meeting of Asia Oceania Geosciences Society (AOGS2015). Singapore. [poster]
- [13] Arcilla, C.A., Codillo, E.A., **Aquino, K.A.**, Leong, J.A., Sunico, R.S., (2012). Natural Zeolite from Pangasinan, Philippines: Thermal Evolution from 30°C to 1000°C: 2nd Asian Clay Conference, Ewha Womans University, Seoul, Korea. [poster]

AFFILIATIONS

2015-present	European Association of Geochemistry
2015-present	Geochemical Society
2015-2018	Asia Oceania Geological Society
2014-2018	Geological Society of the Philippines
2013-present	University of the Philippines Geology Alumni Association
2010-present	UP Cwm Zena'na Faculty Adviser, 2016 to present; President, First Semester AY2013-2014; Treasurer, AY2011-2012, AY2012-2013 Member since 2010; Event Head, Earth Talk 2012 entitled Unconventional Shale Resources: An Energy Source Game-Changer; Finance and Registration Committee Head, 26th Annual Geology Board Exam Review; Finance and Registration Committee Head, 27th Annual Geology Board Exam Review; Overall Head, 28th Annual Geology Board Exam Review
2009-present	UP Pre-Medical Honor Society Member since 2009; Member, Secretarial Committee AY2009-2010, AY2010-2011; Liason, Secretarial Committee AY2011-2012; Sponsorship Committee Head, Finance and Marketing Committee, AY2012-2013

SELECTED FIELD EXPERIENCE

- [1] Baguio District, Philippines. Geologic and structural mapping, as instructor. Annual Geol112 (Structural Geology) class; 2013-2017
- [2] Ilocos Norte, Philippines, Geologic mapping, as instructor. Geol170 (Field Geology) class, June 2017
- [3] Paly Island and Dumararan, Northern Palawan. Geologic mapping; June 2016
- [4] Infanta and Real, Quezon + Western Rizal, Philippines. Geologic mapping, as instructor. Geol170 (Field Geology) class; June 2015
- [5] Bulacan and Western Rizal, Philippines. Geologic mapping; April 6-21, 2013

RELEVANT SKILLS

- Mineral identification via transmitted and reflected light microscopy
- Geologic mapping and field techniques in geology
- Laboratory skills and operations particularly on X-Ray Diffraction (table top and portable), Rietveld refinement using Siroquant software, X-Ray Fluorescence Spectroscopy (table top and portable), Fourier Transform Infrared Spectroscopy and Real Time Kinematics, Confocal Raman Spectrometry, Gas Pycnometer, Scanning Electron Microscopy
- Use of geochemical softwares such as Geochemist's Workbench and PHREEQC
- Skills on mapping and remote sensing softwares such as Global Mapper, MapInfo Professional, ArcGIS, QGIS, ENVI and Leapfrog Geo
- Computer-related document processing skills (MS Word, Excel, PowerPoint, Google Documents, Inkscape, Adobe Illustrator)
- Leadership and organizational skills, written and oral communication skills, technical writing